A Classical Reasoning Agent for RuleTaker, CounterBench and FOLIO

An AI Assignment Report

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Abstract

This report presents an end-to-end system that determines whether a query is entailed by a natural-language knowledge base containing facts and rules. A rule-based parser converts text into an internal logical representation, and a router selects one of three classical reasoning engines: stratified forward chaining with unification for RuleTaker, structural causal model inference for CounterBench, and resolution refutation for FOLIO. No generative or learned model is used to produce the final True/False decision.

Index Terms— Classical reasoning, RuleTaker, CounterBench, FOLIO, forward chaining, unification, structural causal models, resolution.

1 System Overview

The agent consists of three classical reasoning engines and a router that selects an engine from the structure of an input. CounterBench is recognised by its causal vocabulary and counterfactual question form, the explicit `Facts:` / `Rules:` format is routed to the propositional RuleTaker engine, and short relational rule contexts are handled by the relational RuleTaker engine. Inputs containing explicit first-order logic notation, and other general logical examples, are routed to FOLIO.

Dataset	Internal representation	Reasoning algorithm
RuleTaker	Horn rules and facts; propositional atoms for the Facts/Rules format	Stratified forward chaining, unification, and NAF
CounterBench	Structural causal model (AND/OR/NOT)	Abduction, intervention, and forward inference
FOLIO	First-order clauses (CNF)	Resolution refutation with unification

2 Chosen Representation

2.1 Literals

Every sentence is parsed using regular expressions, without a learned model, into literals $(\sigma, p, \vec{t})$, where $\sigma \in \{+, -\}$ is the polarity, p is a predicate, and $\vec{t}$ is the argument tuple.

- Unary attributes: "Bob is young" maps to $young(bob)$.
- Binary relations: "The bear chases the dog" maps to $chase(bear, dog)$. Verbs are lemmatised by stripping the trailing s; the determiner "the" is dropped.
- Negation: "is not", "are not", and "does not" set $\sigma = -$.

2.2 Rules and Facts

Sentences are converted into Horn clauses of the form

$$B_1 \wedge \cdots \wedge B_n \rightarrow H.$$

A fact is a rule with an empty body.

The parser supports the following templates:

Natural language	Clause
Bob is young.	$young(bob)$
If someone is young, then they are smart.	$young(X) \rightarrow smart(X)$
If someone is smart and kind, then they are successful.	$smart(X) \wedge kind(X) \rightarrow successful(X)$
Big, red things are kind.	$big(X) \wedge red(X) \rightarrow kind(X)$
Everything is cold.	$\rightarrow cold(X)$
If something chases the dog then it is nice.	$chase(X, dog) \rightarrow nice(X)$

Pronouns and quantifier words (*someone, something, it, they, them*) map to a single rule variable X. Elliptical conjunctions, such as "smart and kind", inherit the subject of the previous conjunct.

2.3 Propositional Format of the Public RuleTaker Data

The `Jise/ruletaker` data lists `Facts:` and `Rules:`, with sentences paraphrased using varying cue phrases such as "Assuming A, we can say B", "If A, then B", "Based on A, it is evident that B", "Given that A, it follows that B", and "When A, we can conclude B".

Each normalised sentence (lower-cased, with articles removed) is treated as a propositional atom. A rule is split at the first comma into body and head after stripping the cue phrases.

These Horn clauses are solved by linear-time forward chaining with premise counters (Dowling–Gallier). Every rule keeps the number of unsatisfied body atoms and fires when the count reaches zero. Queries are True if and only if the atom is derived under the closed-world assumption. Rule lines that the parser cannot split are counted and reported.

2.4 Two Semantics

Closed-world (CWA). A negative body literal $\neg p(\vec{t})$ is true if $p(\vec{t})$ is not derivable (negation as failure). Rules with negative heads are ignored.

Open-world (OWA). Negation is explicit: $\neg p$ is stored as a separate predicate $\sim p$, which is true only if asserted or derived. An unknown positive or negative statement is not assumed true; the system's required binary output reports it as False.

The public `Jise/ruletaker` data, in the propositional `Facts:/Rules:` format, is evaluated under the closed-world reading. The OWA variant is implemented for the relational engine.

3 Reasoning Algorithms

3.1 Forward Chaining with Unification

Starting from the facts, each rule body is matched against the current fact base by a left-to-right join. Matching uses *unification*: a rule atom such as $young(X)$ is unified with a ground fact $young(bob)$, extending the substitution $\{X \mapsto bob\}$.

This substitution is propagated to the remaining body atoms. When all body atoms match, the instantiated head is added. The process repeats until no new fact is produced, reaching a fixpoint. Variables occurring only in the head or in negated literals are grounded over the entity set of the knowledge base.

3.2 Stratification for Negation as Failure

Under CWA, the result of negation as failure depends on evaluation order unless the program is stratified.

A level $\ell(\cdot)$ is computed for each predicate such that

$$\ell(h) \geq \ell(b) \quad \text{for positive } b \in \text{body},$$

$$\ell(h) > \ell(b) \quad \text{for negated } b \in \text{body},$$

using iterative relaxation.

Strata are evaluated in increasing order, each to a fixpoint, so that every predicate used under $\neg$ is complete before it is tested.

3.3 Query Answering and Complexity

A positive query is True if and only if its atom is in the computed model. A negated query is True if the atom is absent under CWA or if $\sim p$ is derived under OWA; otherwise, the answer is False.

The model of a context is computed once and cached, so further queries on the same context cost a set lookup. This addresses the efficiency criterion of total execution time, since RuleTaker contexts contain several questions each.

For a context with E entities and a single rule variable, the model has at most

$$O(P \cdot E^a)$$

atoms, where P is the number of predicates and the arity $a \leq 2$. Each fixpoint round is polynomial in this size.

3.4 Counterfactual Inference (CounterBench)

1. *Abduction:* Starting from the observed values, requirements are propagated backwards through the equations. An AND that is true forces all its parents; an OR that is false forces all its parents; and a single-parent variable forces its parent. Unconstrained root variables default to 0.
2. *Intervention:* The equations of the intervened variables are replaced by constants.
3. *Forward inference:* The value of Y is evaluated by memoised depth-first evaluation of the equations.

The answer is Yes if and only if $Y = 1$. The two defaults (root default and use of the "instead of" factual value) are compared on the public data in the experiments.

3.5 Resolution Refutation (FOLIO)

To test $KB \models \varphi$, the clauses of $\neg\varphi$ are added and the system searches for the empty clause using binary resolution, most general unifiers with occurs check, factoring, and a set-of-support strategy that prefers short clauses. The search is bounded by a time limit.

The predicted label is True if the refutation succeeds. For three-way FOLIO labels, a second refutation of φ itself gives False (contradiction); otherwise, the result is Uncertain. The binary output of the agent maps both False and Uncertain to "not entailed".

3.6 Algorithm Summary

```
parse(context) -> rules, facts
compute strata of predicates
for level in increasing order:
    repeat until no new atom:
        for each rule of this level:
            for each substitution s matching the positive body (unification):
                ground remaining variables; check negated literals (NAF)
                add head(s) to model, store (rule, premises) as provenance
answer(query) = lookup in model (with NAF / explicit negation)
```

3.7 CounterBench: Structural Causal Model

The statement "We know that ..." is parsed into one equation per variable. "A causes B" is represented as $B := A$, "A and B together cause C" as $C := A \wedge B$, and "A or B causes C" as $C := A \vee B$. A leading "not" negates a parent.

Observations ("We observed V") and the factual side of "if not X instead of X" are stored as known values. Questions yield interventions $do(V_i = v_i)$ and a target Y. Nested questions ("Assume not X, ... further suppose not V_4") are treated as joint interventions.

3.8 FOLIO: First-Order Logic

Formulas use the FOLIO syntax $(\forall, \exists, \neg, \wedge, \vee, \oplus, \rightarrow, \leftrightarrow)$.

Each formula is converted to clause form by eliminating $\rightarrow$, $\leftrightarrow$, and $\oplus$; converting to negation normal form; skolemisation; and distribution of $\vee$ over $\wedge$.

When no FOL is supplied, a heuristic pattern-based translator maps English sentences to FOL. Supported patterns include "All/Every/No/Some ...", "If ... then ...", copular and simple verb sentences, and relative clauses with *who/that*. Sentences it cannot translate are skipped.

4 Experiments on the Public Dataset

Evaluation was performed on the three public datasets assigned in the assignment: `Jise/ruletaker` (splits `train`: 1000, `test`: 250, and `ood_test`: 375 examples; columns `context`, `statement`, and `flag`), `CounterBench/CounterBench` (1200 questions of types basic, conditional, joint, nested, and backdoor), and `yale-nlp/FOLIO` (train subset of 300 and validation 203 examples).

No model is trained; the public data were used to design and debug the parsers. The notebook prints accuracy and total reasoning time for every split.

Dataset / split	Setting	Accuracy (%)	Time (s)
RuleTaker train (1000)	Forward chaining	98.30	0.20
RuleTaker test (250)	Forward chaining	97.60	0.05
RuleTaker ood_test (375)	Forward chaining	99.47	0.07
CounterBench (1200)	SCM, abduction including "instead of" (default)	94.58	0.234
CounterBench (1200)	Ablation: no "instead of" abduction	91.67	0.357
CounterBench (1200)	Ablation: unknown roots = 1	95.00	0.152
FOLIO train subset (300)	Resolution, gold FOL	79.33	33.57
FOLIO train subset (300)	Resolution, NL→FOL	62.67	0.28
FOLIO validation (203)	Resolution, gold FOL	81.28	62.28
FOLIO validation (203)	Resolution, NL→FOL	63.55	0.12

FOLIO accuracies are binary (entailed versus not entailed). With gold FOL, the three-way accuracy (True/False/Uncertain) is 67.0% on validation. For the FOLIO train subset, the reported results are 79.3% binary and 62.3% three-way accuracy with gold FOL, and 62.7% binary accuracy with NL→FOL.

CounterBench accuracy by question type under the default settings is: basic 100%, conditional 90%, joint 96%, nested 96%, and backdoor 90%.

Error Analysis. *RuleTaker:* All rule lines were parsed (0 unparsed), and accuracy is 97.6–99.5%. The remaining errors come from statements whose normalised atom differs slightly from atoms in the facts and rules, or from negated statements handled by negation as failure.

CounterBench: The parser handles negative causal effects such as "Trune causes *not* Vork" by applying De Morgan's law: the AND/OR operator is flipped and the polarity of each parent input is inverted. The current evaluation reaches 94.58% overall accuracy, with basic questions at 100%. The dataset inspection cell identifies 580 contexts containing negative causal effects and 680 contexts containing joint causal mechanisms.

The ablation shows that using the factual value implied by "if not X instead of X" matters mostly for backdoor questions: backdoor accuracy falls from 90% to 72.5% when this information is not used. Changing the default value of unconstrained root variables from 0 to 1 changes the overall accuracy from 94.58% to 95.00%. Remaining errors occur mainly in conditional and backdoor questions, where observations with OR-gates can leave several possible causes and the implemented abduction procedure does not branch.

FOLIO: With gold FOL, the resolution prover reaches 81.3% binary accuracy, so most of the loss on raw text is attributed to the translation step. The heuristic translator reaches 63.6% because many FOLIO sentences fall outside its patterns and are skipped. The label distribution was not measured, so comparison with a constant "not entailed" answer cannot be made.

The gold-FOL validation run takes 62.28 s for 203 examples. The prover uses a bounded resolution search; a query that cannot be settled within the search budget may be returned as not entailed by the binary prediction interface. Gold FOL is useful for measuring the prover separately from the translator, but the natural-language path is the relevant path when logical annotations are unavailable. The raw-text validation run takes 0.12 s.

Sanity Tests. Hand-written theories (the assignment example, chains with negation as failure, multi-premise rules, relations, and OWA negation) were all answered correctly.

4.1 Submission CSV

The notebook's final submission cell is configured with `TEST_PATH = None` for a reproducible public demonstration. In this mode, it uses the public RuleTaker test split and writes `submission.csv` with the columns `id` and `label`. The saved run contains 250 predictions, reports zero processing errors, and obtains 97.60% public RuleTaker test accuracy. This file is a demonstration artifact, not a prediction file for the instructor's

private test set. When an evaluation CSV is provided by the instructor, its path can be assigned to `TEST_PATH` and the notebook can be rerun to generate predictions for that input. No private test data are used during development.

5 Examples of Reasoning Traces

Example 1 (assignment example), query: *Bob is successful* ⇒ True.

```
successful(bob) <= "If someone is smart and kind, then they are successful."
  smart(bob) <= "If someone is young, then they are smart."
    young(bob) [fact]
  kind(bob) [fact]
```

Example 2 (negation as failure), query: *The bear is kind* ⇒ True. Context: *The bear is big. The bear chases the dog. If something chases the dog then it is nice. If something is nice and not green then the dog sees it. If the dog sees the bear then the bear is kind.*

```
kind(bear) <= "If the dog sees the bear then the bear is kind."
  see(dog, bear) <= "If something is nice and not green then the dog sees it."
    nice(bear) <= "If something chases the dog then it is nice."
      chase(bear, dog) [fact]
    not green(bear) [negation as failure]
```

Example 3 (CounterBench), query: *Would Wrox occur if not Nuv and not Splee?* ⇒ Yes. The context states that Nuv causes Splee, Blen, and Druk; not Druk causes Plog; Plog causes Skrim; Skrim or Druk causes Zimb; and Zimb causes Yurd, which causes Wrox. The question sets $do(Nuv = 0, Splee = 0)$. With the default root value 0, Druk is 0, so Plog becomes 1, followed by Skrim = 1, Zimb = 1, Yurd = 1, and Wrox = 1. The target is therefore True.

Example 4 (FOLIO, resolution). Premises:

$$\forall x\big(Human(x) \to Mortal(x)\big), \qquad Human(socrates).$$

Conclusion:

$$Mortal(socrates).$$

The clauses are $\{\neg H(x), M(x)\}$, $\{H(s)\}$, and the negated goal $\{\neg M(s)\}$.

Resolving $\{\neg M(s)\}$ with $\{\neg H(x), M(x)\}$ using $x \mapsto s$ gives $\{\neg H(s)\}$. This resolves with $\{H(s)\}$ to produce the empty clause, establishing that the conclusion is True.

Example 5 (False answer). The query is *Bob is rich*, using the Example 1 context. No rule has *rich* in its head, and no such fact exists. Therefore, the atom is absent from the model and the answer is False.

6 Conclusion

The developed agent combines three classical reasoning engines: stratified forward chaining with unification, structural-causal-model inference with abduction and intervention, and first-order resolution refutation. On the reported public experiments, it achieves 97.60–99.47% accuracy across the RuleTaker splits, 94.58% on CounterBench with the default settings (95.00% when the root default is set to 1), and 81.28% on FOLIO validation when gold logical forms are provided. The natural-language-to-FOL validation accuracy is 63.55%, reflecting the limitations of the pattern-based translator.

The main open problem is open-domain English-to-FOL translation for FOLIO.

7 Limitations

The RuleTaker parser targets the RuleTaker templates. The CounterBench semantics (abduction rules and default root value) were designed from the dataset description and tuned on the public data.

The English-to-FOL translator for FOLIO is a shallow, pattern-based baseline. FOLIO sentences are open-domain and often fall outside its patterns, so accuracy without gold FOL is considerably lower than that of the prover on gold FOL.

Unusual phrasings, multi-word entity names used as objects, and rules with more than one variable may be mis-parsed. Closed-world reasoning is assumed for the Facts/Rules format.